

SOUTHWEST MINNESOTA, MN, USA

WMO: 726559

Lat: 44.450N

Lon: 95.817W

Elev: 1179

StdP: 14.08

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94976

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-8.3	-21.9	1.7	-10.3	-17.1	2.3	-7.3	36.0	11.3	32.9	15.3	11.0	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	90.3	73.3	87.9	72.6	84.1	70.3	76.9	86.6	74.8	83.9	72.8	81.6	15.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	129.5	83.1	72.0	123.8	81.8	69.9	115.1	79.7	41.2	86.6	39.1	83.3	37.2	81.3	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
31.1	27.1	24.6	DB	-17.8	95.8	5.4	2.8	-21.6	97.8	-24.8	99.5	-27.8	101.0	-31.7	103.1	
			WB	-18.0	80.0	4.9	2.2	-21.5	81.6	-24.4	82.9	-27.2	84.2	-30.7	85.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	15.4	19.3	31.5	45.0	57.8	68.1	72.2	69.4	62.3	48.6	33.8	19.8
	DBStd	22.59	13.54	13.36	13.18	10.44	9.26	7.06	6.11	5.72	8.82	9.81	11.66	12.50
	HDD50	4349	1071	859	585	213	33	0	0	0	9	145	498	936
	HDD65	7804	1536	1279	1038	603	260	47	11	23	155	515	936	1401
	CDD50	2674	0	0	13	63	275	543	689	600	377	102	12	0
	CDD65	654	0	0	1	3	37	140	235	158	73	7	0	0
	CDH74	6148	0	0	4	55	445	1356	2252	1279	673	83	1	0
	CDH80	1949	0	0	0	13	147	446	785	347	193	18	0	0
Wind	WSAvg	11.7	12.6	13.0	12.7	13.1	12.8	10.9	9.2	8.7	10.4	12.1	12.6	12.4
Precipitation	PrecAvg	28.1	0.8	0.7	1.5	2.8	3.4	4.4	3.3	3.6	3.2	2.3	1.3	0.8
	PrecMax	39.6	2.1	1.5	3.5	7.5	6.7	11.0	6.2	6.7	12.1	6.5	4.2	2.3
	PrecMin	21.9	0.0	0.0	0.2	0.3	1.0	1.7	0.4	0.9	0.6	0.2	0.0	0.0
	PrecStd	4.9	0.6	0.4	0.9	1.5	1.7	2.0	1.5	1.6	2.6	1.7	1.3	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	55.7	71.8	81.2	90.8	92.8	94.7	90.6	90.2	82.2	69.7	53.6
		MCWB	39.1	48.2	55.6	60.7	66.7	71.2	78.0	76.2	71.2	62.1	53.9	45.6
	2%	DB	40.1	46.4	62.9	73.0	83.8	88.6	90.5	87.8	85.0	74.9	61.7	44.9
		MCWB	35.0	40.6	52.8	55.9	64.4	70.1	75.6	75.2	69.9	59.1	50.0	38.0
	5%	DB	36.8	41.3	55.1	67.5	79.1	84.4	87.9	83.7	81.5	69.9	55.2	39.1
		MCWB	33.2	36.2	47.5	53.0	61.4	68.0	74.7	72.8	66.8	56.7	46.3	34.3
	10%	DB	34.2	36.9	49.5	62.9	73.3	81.6	84.0	81.2	77.5	64.0	50.2	36.1
		MCWB	31.5	33.4	43.9	50.3	58.6	66.0	72.3	69.9	64.6	53.3	42.8	32.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.8	47.8	59.1	62.2	71.2	76.5	80.8	79.0	74.4	66.3	57.0	45.9
		MCDB	46.4	55.9	65.9	77.3	85.4	86.4	89.4	87.6	84.9	77.4	65.2	51.4
	2%	WB	35.7	40.9	53.4	58.2	67.1	73.3	77.8	76.1	71.8	61.6	51.6	38.4
		MCDB	39.7	46.7	60.0	70.1	79.5	84.2	88.1	84.7	81.7	71.3	59.7	43.6
	5%	WB	33.5	36.9	48.8	54.6	64.0	70.6	75.6	74.0	69.3	57.9	47.0	34.9
		MCDB	36.9	41.2	55.9	64.9	75.6	82.0	85.0	81.8	78.3	67.5	55.6	39.3
	10%	WB	31.4	33.6	43.6	51.5	60.9	68.3	73.6	71.5	66.8	54.3	42.8	32.4
		MCDB	33.9	37.2	49.2	61.0	70.5	79.1	82.1	79.1	75.3	63.1	49.6	35.7
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.6	16.5	19.7	20.3	19.3	19.4	19.3	21.6	19.5	16.6	14.9
		MCDBR	18.6	20.2	24.4	29.9	27.7	23.6	21.3	21.4	24.8	26.8	24.5	18.9
		MCWBR	15.1	15.2	16.5	17.2	14.3	11.6	11.4	12.1	12.9	15.1	16.3	14.6
	5% WB	MCDBR	17.9	19.5	23.7	25.6	23.4	21.3	19.9	19.6	21.2	23.4	23.0	18.0
		MCWBR	14.7	15.0	16.5	16.0	13.5	12.0	11.6	12.2	12.5	14.5	15.8	14.2
Clear-Sky Solar Irradiance	taub	0.263	0.276	0.309	0.362	0.381	0.395	0.404	0.408	0.365	0.322	0.286	0.267	
	taud	2.455	2.430	2.438	2.328	2.344	2.364	2.324	2.290	2.388	2.493	2.516	2.456	
	Ebn at Noon	278	293	295	285	282	278	273	268	272	272	267	264	
	Edh at Noon	22	27	31	38	39	38	39	39	33	26	21	20	
All-Sky Solar Radiation	RadAvg	590	920	1209	1507	1741	1938	2097	1726	1354	892	590	443	
	RadStd	63	105	123	151	128	134	72	141	101	115	51	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	-1.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page